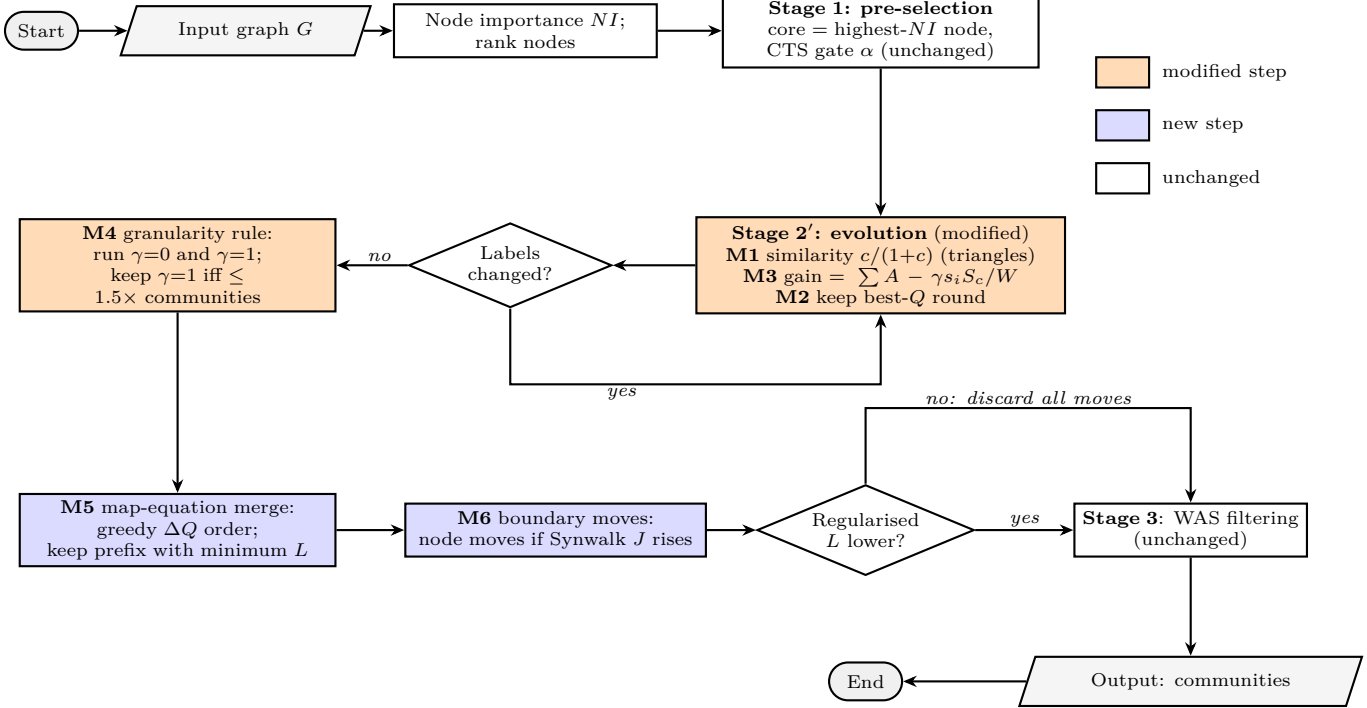
Start
Input graph G
Node importance NI; rank nodes
Stage 1: pre-selection
core = highest-NI node,
CTS gate α (unchanged)
modified step
new step
unchanged
Stage 2′: evolution (modified)
M1 similarity $c/(1+c)$ (triangles)
M3 gain = $\sum A - \gamma s_i S_c/W$
M2 keep best-Q round
Labels changed?
no
yes
M4 granularity rule:
run γ=0 and γ=1;
keep γ=1 iff $\leq$
1.5× communities
M5 map-equation merge:
greedy ΔQ order;
keep prefix with minimum L
M6 boundary moves:
node moves if Synwalk J rises
Regularised L lower?
yes
no: discard all moves
Stage 3: WAS filtering
(unchanged)
Output: communities
End